

Aether — Cost Analysis

Multi-Cloud Cost Estimation & Comparison

Estimate infrastructure costs across five cloud providers before deploying.
Compare runtimes side by side and make cost-aware scheduling decisions
from the CLI.

AWS • Azure • GCP • DigitalOcean • Linode

Supported Providers

Aether includes built-in pricing models for five major cloud providers, with per-resource cost breakdowns.



AWS

EC2 instance pricing based on vCPU and memory. Includes EBS storage, network egress, and ALB load balancer costs. Based on us-east-1 on-demand rates.

- CPU: \$15.00/core/month
- Memory: \$4.00/GB/month
- Storage: \$0.10/GB/month
- GPU: \$350.00/unit/month



Azure

Virtual Machine pricing with Azure Disk Storage and bandwidth. Based on East US standard tier pricing.

- CPU: \$14.00/core/month
- Memory: \$3.50/GB/month
- Storage: \$0.12/GB/month
- GPU: \$320.00/unit/month



GCP

Compute Engine pricing with Persistent Disk and network costs. Based on us-central1 on-demand rates.

- CPU: \$13.50/core/month
- Memory: \$3.75/GB/month
- Storage: \$0.08/GB/month
- GPU: \$300.00/unit/month



DigitalOcean & Linode

Droplet and Linode pricing with block storage. Simpler pricing models with flat per-resource rates.

- DO CPU: \$12.00/core/month
- DO Memory: \$3.00/GB/month
- Linode CPU: \$11.00/core/month
- Linode Memory: \$2.50/GB/month

Cost Methodology

Aether reads resource requirements from your workload spec and applies provider-specific rates to calculate monthly and hourly cost estimates.



Cost Breakdown Structure

CostEstimate Fields

Field	Description
provider	Cloud provider name
cpu_cost_monthly	Monthly cost for CPU cores
memory_cost_monthly	Monthly cost for RAM
storage_cost_monthly	Monthly cost for disk storage
total_monthly	Sum of all monthly costs
total_hourly	total_monthly / 730 hours
currency	Always USD

Resource Parsing

```
# Your workload spec
requirements:
  cpu: "4"           # 4 cores
  memory: 16Gi       # 16 GB RAM
  storage: 100Gi     # 100 GB disk
  gpu:
    count: 1
    vendor: nvidia

# Aether calculates:
# CPU:      4 cores x $15.00 = $60.00/mo
# Memory:   16 GB   x $4.00  = $64.00/mo
# Storage:  100 GB  x $0.10   = $10.00/mo
# GPU:      1 unit  x $350.00 = $350.00/mo
# ---
# Total: $484.00/mo ($0.6630/hr)
```

CLI Usage

Estimate costs for a single provider, compare all providers, or compare across runtimes.

Single Provider

```
$ aether cost workload.yaml --provider aws
```

AWS Cost Estimate for my-app:

```
CPU:      $60.00/mo (4 cores)
Memory:   $64.00/mo (16 GB)
Storage:  $10.00/mo (100 GB)
GPU:      $350.00/mo (1x NVIDIA)
-----
Total:    $484.00/mo ($0.6630/hr)
```

All Providers

```
$ aether cost workload.yaml --provider all
```

Provider	Monthly	Hourly	Cheapest
Linode	\$394.50	\$0.5404	
DigitalOcean	\$408.00	\$0.5589	
GCP	\$442.80	\$0.6066	
Azure	\$467.20	\$0.6400	
AWS	\$484.00	\$0.6630	

Cheapest: Linode (\$394.50/mo)

Savings vs most expensive: \$89.50/mo (18.5%)

Cross-Runtime Comparison

```
$ aether compare workload.yaml
```

Runtime Comparison for my-app (4 CPU, 16Gi RAM, 100Gi storage):

Runtime	Best Provider	Monthly	Startup	Isolation	Migration
Podman	Local	\$0.00	Seconds	Container	Immediate
Kubernetes	Linode	\$394.50	Seconds	Container+NP	Blue-green
KubeVirt	GCP	\$442.80	Minutes	Full VM	Blue-green
Metal3	On-prem	Custom	Minutes	Physical	Rolling

Recommendation: Start with Podman for dev, deploy to Kubernetes on Linode for production.

Cost-Aware Scheduling

Aether's Decision Engine can factor cost into runtime selection, choosing the most economical target that meets your workload requirements.



Automatic Cost Optimization

When the runtime preference is set to `auto`, the Decision Engine evaluates cost alongside isolation requirements, scaling needs, and hardware constraints to select the optimal runtime.



Budget Constraints

Set maximum monthly budgets per workload. Aether will warn when a deployment would exceed the budget and suggest cheaper alternatives.



Scaling Cost Projection

When using HPA, Aether projects costs at minimum, target, and maximum replica counts so you can understand the cost envelope before deploying.



JSON Output for Analysis

Export cost estimates as JSON for integration with financial dashboards, FinOps tools, and custom reporting pipelines.

Cost Projection with Scaling

```
$ aether cost workload.yaml --provider aws --with-scaling
```

AWS Cost Estimate for my-app (HPA: 2-10 replicas):

Replica Count	Monthly	Hourly
2 (min)	\$268.00	\$0.3671
5 (target)	\$670.00	\$0.9178
10 (max)	\$1,340.00	\$1.8356

Per-replica: \$134.00/mo

Tip: At 70% CPU utilization target, expect 3-5 replicas under typical load (\$402-\$670/mo).



Cost estimation is not a deployment blocker — it is a decision-support tool. Aether provides the data; your team makes the call.

Know Your Costs **Before Deploy**

Estimate, compare, and optimize infrastructure costs across five cloud providers and four runtimes.

Quick Cost Check

```
# Estimate cost for your workload
$ aether cost workload.yaml --provider all --output json

# Compare runtimes side by side
$ aether compare workload.yaml

# Project scaling costs
$ aether cost workload.yaml --provider aws --with-scaling
```

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Cloud Providers

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Cost Dimensions

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Runtimes Compared

USD

Currency

FinOps Built Into Your Deployment Tool

Aether puts cost visibility directly in the developer workflow. No separate FinOps tool, no manual spreadsheets — just run `aether cost` and see exactly what your workload will cost across every provider.